together define the state of a classical system. Indeed, in the Hamiltonian formulation of classical mechanics, position and momentum are explicitly on an equal footing. Is this a reflection of some underlying symmetry that isn't obvious on the surface?

In our everyday lives, position and momentum seem quite different. What a mathematician would call "the space of all possible positions" is what the rest of us just call "space"; it's the three-dimensional world in which we live. The "space of all possible momenta," or "momentum space," is also three-dimensional, but it's a seemingly abstract concept. Nobody believes we live there. Why not?

The feature that makes space special is locality. Interactions between different objects happen when they are nearby in space. Two billiard balls bounce off each other when they come together at the same spatial position. Nothing of the sort happens when particles have the same (or opposite) momenta; if they're not in the same location, they just keep going their merry way. That's not a necessary feature of the laws of physics—we could imagine other possible worlds where it wasn't the case—but it's one that seems to hold pretty well in our world.

Ricocheting billiard balls are classical, but the same discussion could be had about quantum mechanics. The basic quantum formalism also treats position and momentum equally. We can express the wave function by attaching a complex amplitude to every possible location the particle can be in, or we could just as well express it by attaching a complex number to every possible momentum the particle could have. The two ways of describing the same underlying quantum state are equivalent, expressing the same information in different ways, as we saw when discussing the uncertainty principle.

This is kind of profound. We've said that a wave function of definite momentum looks like a sine wave. But that's what it looks like in terms of position, which is the language we naturally tend to speak. Expressed in terms of momentum, the same quantum state would look like a spike located at that particular momentum. A state with definite position would look like a sine wave spread over all possible momenta. This begins to suggest that what really matters is the abstract notion of "the quantum state," not its specific realization as a wave function in terms of either position or momentum.